Sahil S. Sharma

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github.com/SharmaSahil2003 | portfolio

Summary

Software engineer with a strong foundation in data structures, algorithms, and full-stack development. Builds production REST APIs and frontend systems at enterprise scale, backed by 1500+ LeetCode problems solved, two published IEEE papers, and a hackathon-winning automated QA pipeline. Seeking a Software Development Engineer role on a high-scale engineering team.

Education

Pune Institute of Computer Technology May 2025
Bachelor of Computer Engineering – CGPA: 9.22/10.00 *Pune, Maharashtra*

SSH Junior College 2021
HSC (XII) – 88.33% *Navi Mumbai, Maharashtra*

Ryan International School 2019
CBSE (X) – 91.80% *Navi Mumbai, Maharashtra*

Experience

ZS Associates June 2025 – Present
Business Technology Solutions Associate *Pune, Maharashtra*

- **Testra – AI-Powered Test Management & Automation Platform:** Developed and scaled React/TypeScript frontend features and REST APIs for test case management, generation, execution, test suites, releases, and UI/ETL/CRM/API testing workflows.
- Built backend APIs for large-scale test case import and bulk operations using batch and parallel processing, improving throughput and scalability for high-volume test data processing.
- Implemented test case generation workflows with real-time progress tracking, agent reasoning logs, workflow monitoring, recovery mechanisms, synchronization, and long-running operation handling.
- Developed platform-wide features including bulk test case updates, grid views, drag-and-drop test suite reordering, Playwright script validation, import/export workflows, execution enhancements, and expandable multi-step test case management.
- Contributed to **Testra 2.0 modernization** by developing reusable React components, responsive layouts, navigation, breadcrumbs, themes, loaders, data tables, and modules for projects, users, releases, audit history, defects, automation suites, skills, reporting, and configuration.
- Integrated backend APIs across project management, user management, releases, document management, reporting, configuration, test suites, test execution, and test case management; optimized API usage, reduced redundant requests, and improved application reliability.
- **Ask Testra – AI-Powered Test Assistant:** Integrated chatbot APIs, document management, workflow orchestration, test case generation, conversation management, reasoning visualization, progress tracking, and stop-generation capabilities into the platform.
- **Business Rule Management System (BRMS):** Developed React frontend modules and REST APIs for creating, editing, cloning, and managing data quality rules and execution jobs, including versioning, audit history, approval workflows, lifecycle management, and environment-based promotion.
- **ContractIQ – Contract Intelligence Platform:** Developed advanced search, filtering, repository management, and frontend enhancements for a contract analytics platform; contributed to RBAC implementation and access-control workflows.

Projects

Agentic Social Media Intelligence Platform | *Python, LLMs, RAG, NLP, Flask, MongoDB, React*

- Built an agentic AI platform combining LLM-based RAG, semantic embeddings, comment clustering, and sentiment analysis to automate social media comment understanding and creator engagement.
- Designed an agent workflow for semantic retrieval, FAQ answering, summarization, and sentiment analysis using Sentence Transformers, cosine similarity, DBSCAN/K-Means, and ML classifiers.
- Engineered Flask REST APIs and MongoDB persistence for comments, embeddings, clusters, sentiment scores, and conversation context, with a React interface for chatbot interactions and analytics.

PortfolioPilot – AI Portfolio Evolution Platform | *React, Python, LLMs, RAG, FastAPI, PostgreSQL, Docker*

- Developing an agentic platform that transforms updated resumes into evolving developer portfolios by extracting structured professional data and synchronizing portfolio content and frontend code.
- Designed a three-level architecture spanning data synchronization, page-level code generation, and full-workspace modification, minimizing unnecessary LLM processing through incremental updates.
- Building versioned AI-generated workspaces with code diffs, rollback, human approval, and automated type-check/build validation with iterative code repair before deployment.

Technical Skills

- **Languages:** C++, Python, Java, TypeScript, JavaScript, SQL
- **Frameworks & Technologies:** React, FastAPI, Flask, Spring Boot, REST APIs, HTML/CSS, Bootstrap
- **Databases & Infrastructure:** PostgreSQL, MySQL, MongoDB, Docker, Git
- **AI/ML:** LLMs, RAG, NLP, Sentence Transformers, Semantic Search, Machine Learning
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems

Achievements

- **1st Place, Takeda AI Hackathon (ZS Associates):** Built ATLAS, an automated QA pipeline validating Tableau dashboards against Databricks data, cutting dashboard-validation turnaround by 30–40%.
- **Competitive Programming:** Solved 1500+ LeetCode problems; maintained 1000-day, 500-day, and 365-day solving streaks.
- **Publications:** *Optimizing User-Content Creator Engagement via Semantic Comment Clustering, Automated Chatbots, and Sentiment Analysis* and *Semantic Comment Analysis for Engagement Optimization* – both published in IEEE.
- **Redington Scholarship** recipient for academic merit and education funding.
- Ranked 47,807 in JEE MAIN 2021; 99.63 percentile in MHT-CET 2021.